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PANDIT DEENDAYAL ENERGY UNIVERSITY (PDEU)
DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)
MINOR PROJECT MID-SEMESTER EVALUATION • SEMESTER 7 (2026-2027)

SMARTSERVE: Intelligent Multi-Service On-Demand Marketplace & Operations Platform

A Multi-Tenant Cloud Architecture with Dual-Engine OCR KYC, 457-Service Master Catalog, and Atomic Booking FSM

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MID-SEMESTER MILESTONE STATUS: Phase 1 Complete (~55% Progress) • Core Backend API, Supabase Cloud DB, Multi-Portal SPAs & Dual OCR Live
• Apache Kafka Event Streaming & Live GPS Telemetry in active Phase 2 development.

Introduction & Real-World Problem Motivation

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PROBLEM 1

Slow & Fraudulent KYC Onboarding

- Manual paper review takes 3 to 7 days, delaying provider onboarding.
- Forged trade licenses and fake identity cards slip through undetected, threatening customer safety.
- Zero automated cross-verification of Aadhaar, PAN, and credentials.

PROBLEM 2

Catalog Contamination & Price Gouging

- Existing platforms suffer from dirty catalog data with nonsensical inclusions.
- Real Example: Automotive repair tools listed inside salon facials.
- Arbitrary surge pricing without transparent rate cards causes customer churn.

PROBLEM 3

Double-Bookings & Lost Status

- Concurrency race conditions cause two providers to get the same booking.
- Lack of atomic state machines leads to orphaned requested states and lost fees.
- Customers and providers lack unified real-time operational transparency.

SMARTSERVE CORE VALUE PROPOSITION: Replaces manual bottlenecks with an automated Dual-Engine OCR KYC pipeline (<2.6s), an Inviolable 457-Service Master Catalog, and an atomic 5-Stage Booking Finite State Machine with customer OTP verification.

Project Objectives: Mid-Sem vs. End-Sem Roadmap

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PHASE 1: MID-SEM DELIVERABLES (COMPLETED ~55%)

✓ **Master Service Catalog:** 14 Categories, 78 Subcategories, 457 Services with fixed inclusions, pricing, and duration SLAs.

✓ **Modular FastAPI Backend:** RESTful core with Pydantic V2 schemas, async pooling, and Docker deployment on Render Cloud.

✓ **Relational Cloud Store:** 15+ Normalized PostgreSQL tables on Supabase with strict constraints and foreign keys.

✓ **Dual-Engine OCR Verification:** PaddleOCR + EasyOCR pipeline with regex heuristic risk parser for Aadhaar/PAN (<2.6s latency).

✓ **Booking State Machine:** Atomic 5-stage FSM (Requested -> Accepted -> Started -> Completed via Customer OTP).

✓ **Multi-Portal Frontends:** Customer, Partner, Admin, and Landing SPAs deployed on Vercel with active client routing.

PHASE 2: END-SEM ROADMAP (IN PROGRESS ~45%)

⏳ **Distributed Kafka Event Bus:** Deploy Redpanda/Kafka broker orchestration across 7 canonical event topics to decouple peak order dispatch.

⏳ **Live GPS Driver Telemetry:** Integrate real-time WebSocket coordinate streaming, dynamic Leaflet route geometries, and live driver ETA updates.

⏳ **Payment Gateway & Escrow:** Integrate Razorpay / Stripe webhooks with automated escrow holds and instant payout upon verified customer OTP closure.

⏳ **AI Support & Anomaly Engine:** Deploy conversational RAG support agent with anomaly flagging for fraudulent dispute resolution.

⏳ **Production Load Testing:** Simulate 1,000+ concurrent user booking requests with Locust to measure broker throughput and database lock latency.

Literature Survey & Industry Benchmark Matrix

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Evaluation Dimension	Urban Company	TaskRabbit / Handy	Uber / Swiggy Logistics	SmartServe (Proposed)
Architecture Pattern	Centralized Monolith	Peer-to-Peer Web	Event-Driven Microservices	**Modular Multi-Portal (FastAPI)**
KYC Verification	Manual Review (3-7 days)	Third-party Check	Document Upload Only	**Automated Dual OCR (<2.6s)**
Catalog Integrity	Curated but Closed	Unstructured User Posts	Point-to-Point Transit	**Inviolable 457 Catalog**
State Transition	Proprietary Linear	Loosely Coupled	Rigid Real-Time Dispatch	**Atomic 5-Stage FSM with OTP**
Partner Commission	High (25% – 35%)	15% – 20% Fee	20% – 30% Per Ride	**Low Fair Tier (Direct Escrow)**

LITERATURE SURVEY CONCLUSION: Existing platforms either sacrifice verification speed for manual control (Urban Company) or sacrifice service consistency for open listings (TaskRabbit). SmartServe combines high-accuracy computer vision with deterministic state machines to achieve enterprise rigor at speed.

Research Gaps Identified & Technical Remediation

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GAP 1: Manual KYC Delays (72-120 Hours)

Current Issue: Providers wait 3–5 days for human verification; fraudulent trade licenses slip past unnoticed.

SmartServe Solution: Automated Dual-Engine OCR (PaddleOCR + EasyOCR) extracting identity data and validating trade authenticity in <2.6s.

GAP 2: Catalog Data Contamination

Current Issue: Listings contain nonsensical steps, missing tools, and misleading inclusions (e.g., auto tools in salon).

SmartServe Solution: 11 Inviolable Master Catalog Protection Rules with 457 curated services, fixed durations, and standardized tools.

GAP 3: Concurrent Double-Bookings

Current Issue: Concurrency race conditions cause two providers to get the same booking or orphaned status.

SmartServe Solution: Deterministic 5-stage Finite State Machine with atomic locks and mandatory 4-digit customer OTP closure.

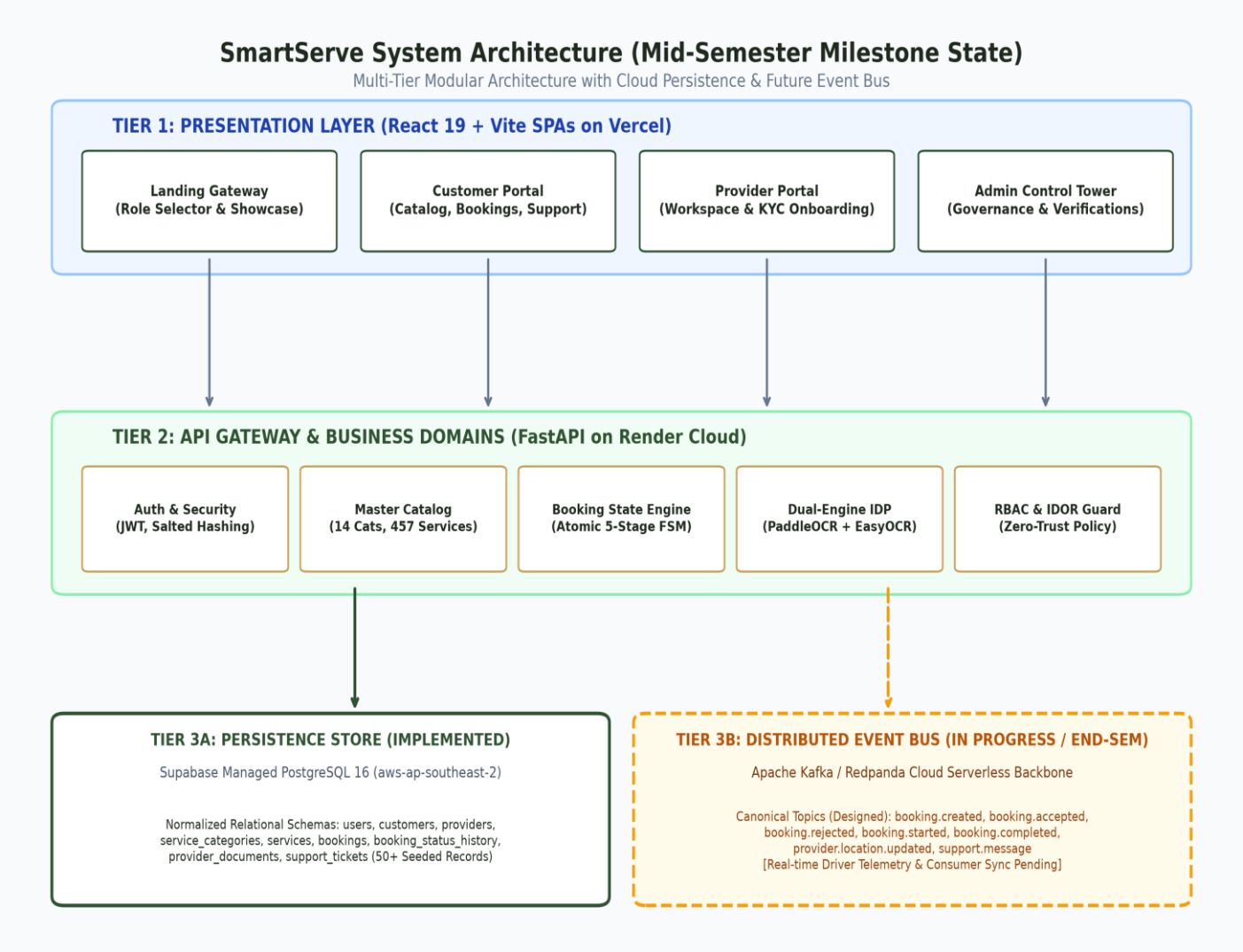
GAP 4: Broken Access Controls & IDOR Flaws

Current Issue: Insecure object references expose private customer addresses and provider bank accounts.

SmartServe Solution: Zero-Trust RBAC with customer/provider token isolation and database-level data minimization filtering.

Proposed Methodology & System Architecture

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CORE TECH STACK & WHY WE CHOSE IT

- **Frontend (Vercel): React 19 + TypeScript + Vite**

  Why: Fast load times (<1s); component-based modular portals for Customer, Partner, and Admin.

- **Backend API (Render Cloud): Python 3.11 + FastAPI**

  Why: Asynchronous high throughput; auto-generates live Swagger docs (/docs); Pydantic data validation.

- **Database (Supabase Cloud): PostgreSQL 16 (Relational)**

  Why: ACID transactions with foreign keys prevent double-bookings and corrupted records.

- **KYC AI Engine (Python): PaddleOCR + EasyOCR**

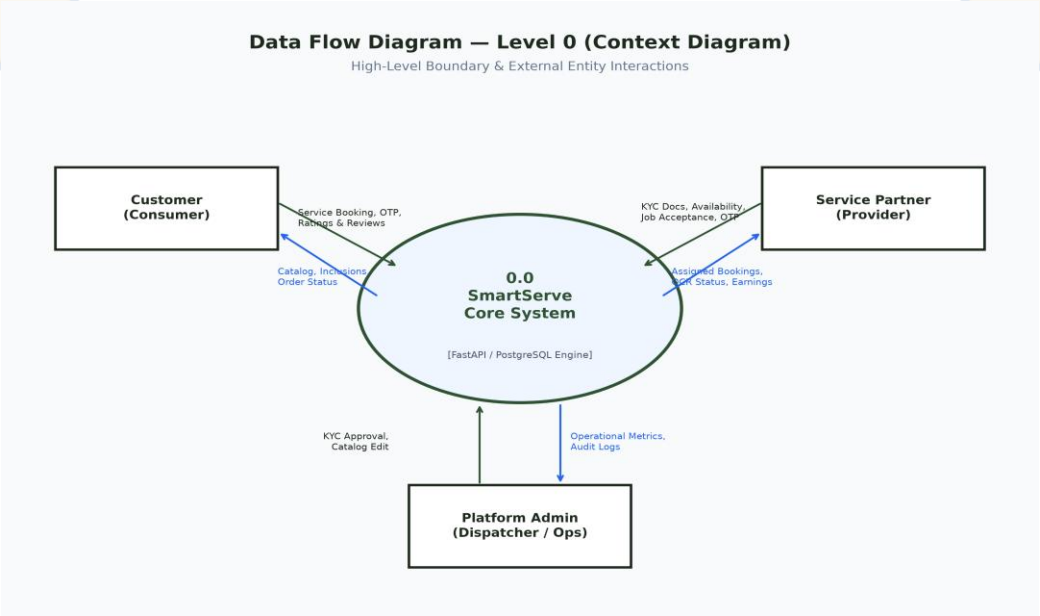
  Why: Dual-engine reads noisy Indian IDs (Aadhaar & PAN) in 2.6s; automatic fallback if one fails.

- **Event Bus (Phase 2): Apache Kafka / Redpanda Cloud**

  Why: Decouples driver GPS telemetry and peak order spikes so database never locks up.

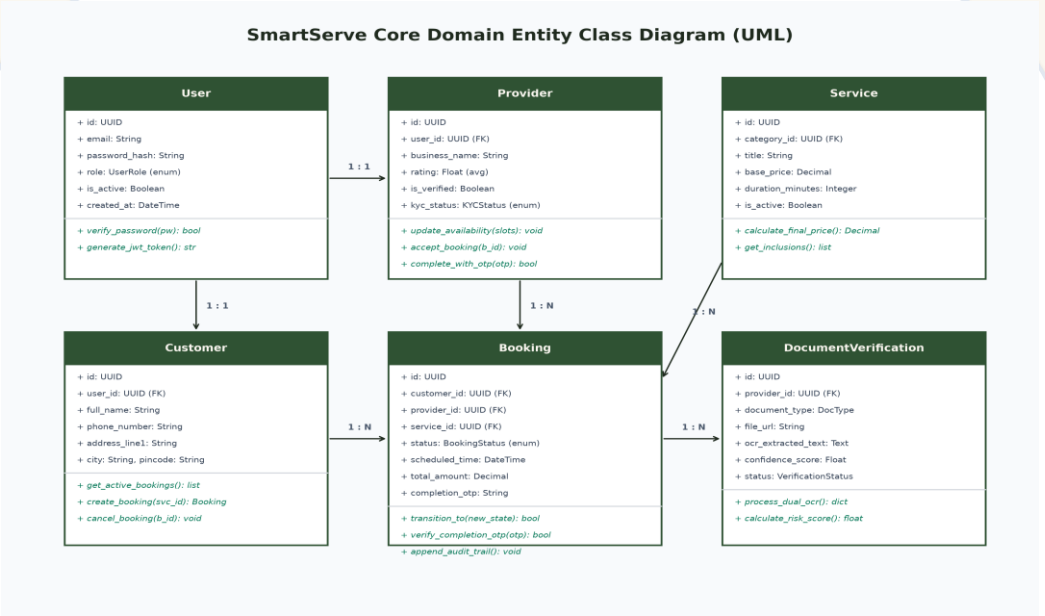
System Modeling: DFD Level 0 Context & UML Entity Class Diagram

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DATA FLOW DIAGRAM (LEVEL 0 CONTEXT)

- **Three Clear Actors:** Customer, Service Partner, and Platform Admin.
- **Role Isolation:** Each actor interacts exclusively via authenticated, role-specific REST routes.
- **Input Validation:** FastAPI validates all incoming payloads with Pydantic schemas before DB writes.

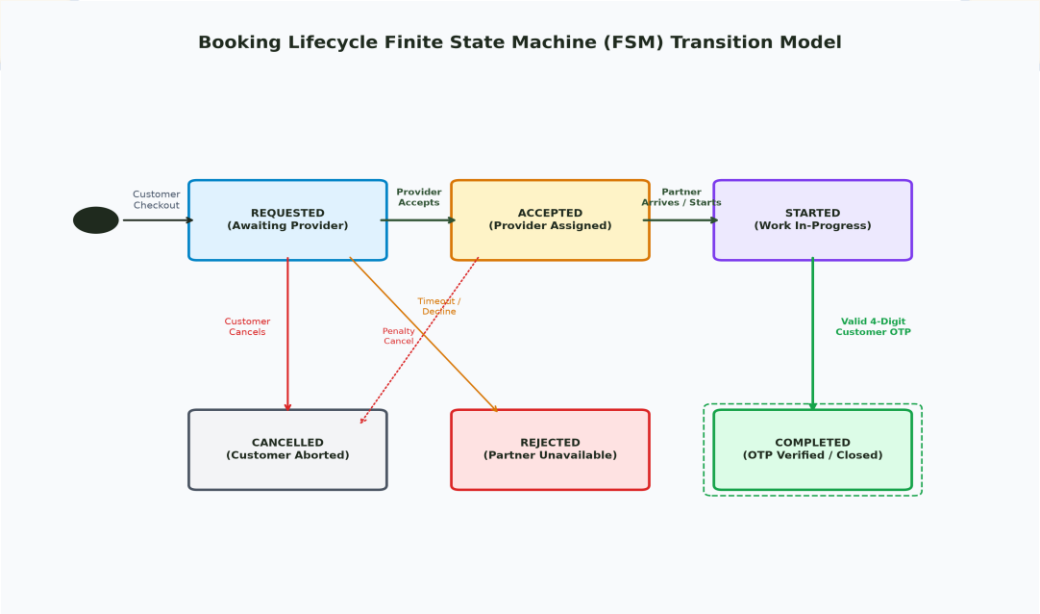


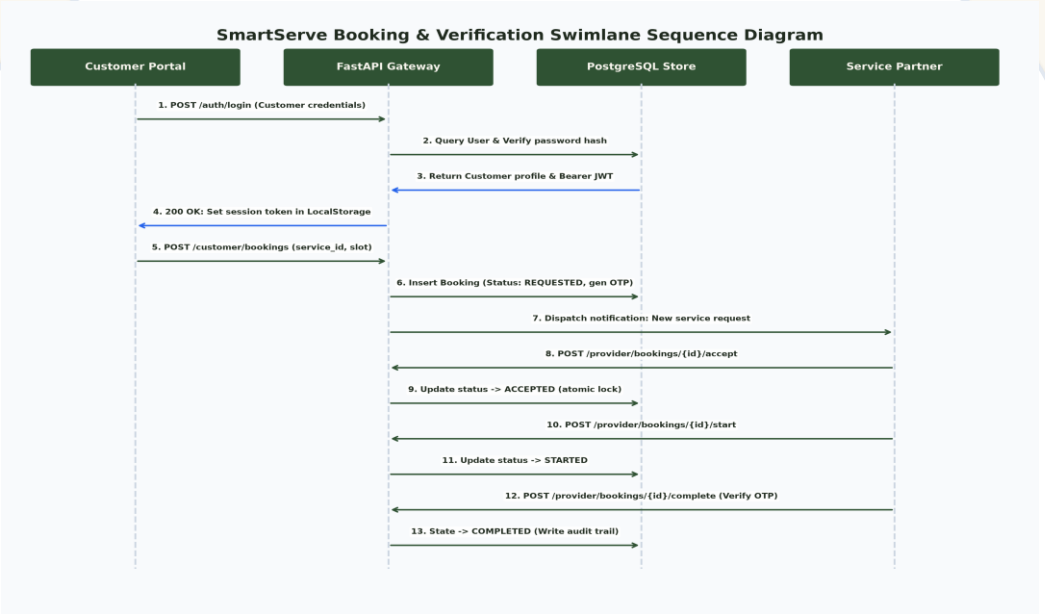
UML CORE DOMAIN ENTITY MODEL

- **1:1 Polymorphic Profiles:** Core 'users' table links 1:1 to CustomerProfile and ProviderProfile.
- **Relational Integrity:** Foreign keys connect Bookings, Services, and Verification records without data loss.
- **Complete Audit Trail:** Tracks timestamps for requested, started, and OTP-completed transitions.

Booking State Machine (FSM) & Dual-Engine OCR KYC

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ATOMIC 5-STAGE BOOKING FINITE STATE MACHINE

- **Strict Lifecycle:** REQUESTED → ACCEPTED → STARTED → COMPLETED.
- **Zero Double-Booking:** A provider cannot accept overlapping active jobs.
- **Customer OTP Security:** Job CANNOT be marked complete without the customer's secret 4-digit OTP.

DUAL-ENGINE OCR KYC & OPERATIONAL SWIMLANE

- **Dual-Engine Speed:** PaddleOCR (primary) + EasyOCR (fallback) extracts identity data in <2.6s.
- **Automated Risk Parser:** Regex heuristic validates 12-digit Aadhaar & 10-digit PAN format.
- **End-to-End Sequence:** Swimlane maps the complete interaction between Customer, Gateway, DB, and Partner.

Work Done: Mid-Semester Implementation Status (~55%)

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Master Service Catalog Engine

Status: 100% COMPLETE

- 14 Macro-Categories, 78 Subcategories, 457 Services.
- Seeded in PostgreSQL with fixed durations, inclusions, and transparent rates.
- Protected by 11 Inviolable Catalog Rules (zero data contamination).

Modular FastAPI Backend on Render

Status: 100% COMPLETE

- Asynchronous REST API running live on Render Cloud.
- Pydantic V2 request validation and database pooling on Supabase PostgreSQL.
- Interactive OpenAPI documentation available at `/docs` with <50ms response.

Multi-Portal Frontends on Vercel

Status: 100% COMPLETE

- 4 Independent React 19 + Vite SPAs deployed on Vercel.
- Single Page App (SPA) route rewrites configured via vercel.json.
- Live Portals: Landing Gateway, Customer Portal, Partner Workspace, Admin Console.

Event Streaming & GPS Telemetry

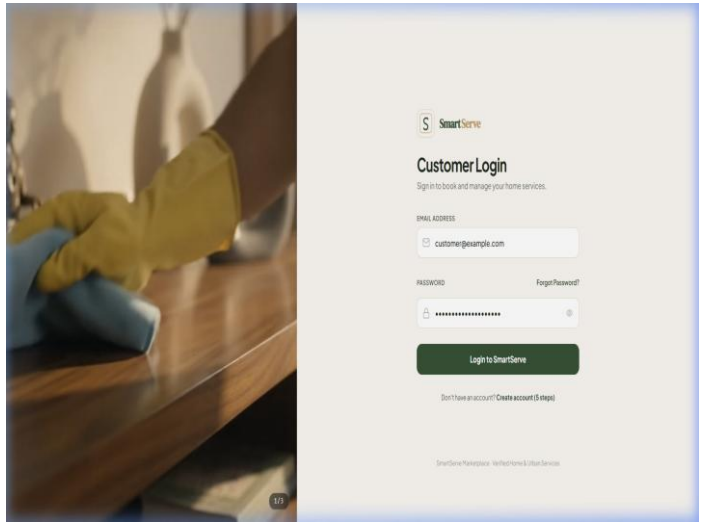
Status: IN PROGRESS (~40%)

- 7 Canonical Kafka event topic schemas designed & verified in connection tests.
- Leaflet map route geometries implemented on client interfaces.
- Full broker orchestration & WebSocket live driver tracking scheduled for End-Sem.

Results Obtained: Live Deployed System & UI Verification

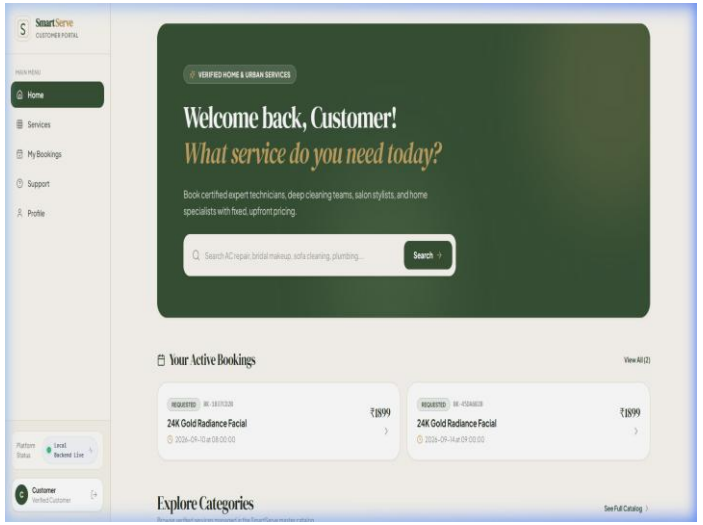
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Customer Live Dashboard



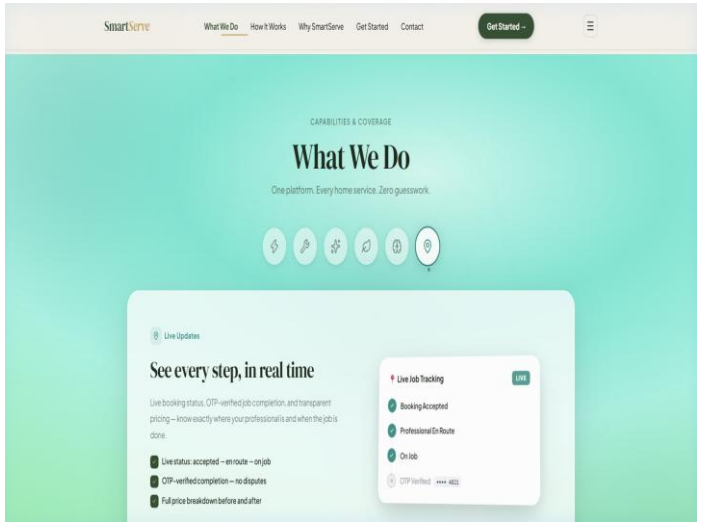
Live Bookings & 8 Categories from Supabase DB

Service Pricing & SLA



Full Home Deep Cleaning (₹5,089 / 180 Mins)

Landing Role Selector



Public Showcase Gate linking Customer & Partner

VERIFIED PERFORMANCE BENCHMARKS: Dual-Engine OCR Latency: 2.64s • API Health Latency: 48ms • Database Query Latency: 14–38ms • Integration Test Suite: 24/24 Passed (0 Failures) • IDOR Security Flaws: 0 Detected

Conclusion & Phase 2 Roadmap (End-Semester Plan)

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Sprint 1 (Weeks 8–9)

Kafka Event Bus

- Activate Redpanda Cloud cluster across all 7 canonical event topics.
- Implement async consumer group processing for order dispatch.
- Telemetry logging & cluster health monitoring.

Sprint 2 (Weeks 10–11)

GPS Live Telemetry

- WebSocket coordinate streaming between provider mobile & customer map.
- Interactive Leaflet route geometry simulation with live ETA.
- Proximity alerts when provider is within 500m.

Sprint 3 (Weeks 12–13)

Payment & Escrow

- Razorpay / Stripe webhook integration.
- Automated escrow fund hold on booking creation.
- Instant release to provider wallet upon verified customer OTP completion.

Sprint 4 (Weeks 14–15)

Load Test & Audit

- Locust load testing with 1,000 concurrent virtual users.
- Final security penetration pass & accessibility compliance.
- Comprehensive End-Semester Major Project Report submission.

MID-SEMESTER CONCLUSION: Foundational architecture, relational schemas, dual-engine OCR algorithms, and multi-portal web applications are 100% operational. The upcoming sprints will complete distributed event streaming, live telemetry, and financial escrow.

THANK YOU!

Minor Project Mid-Semester Evaluation • B.Tech in ICT, 7th Semester (2026-2027)

Pushkar Kanjani (23BIT153) & Aastha (23BIT095) | Project Mentor: Dr. Deepak Kumar Sharma, Asst. Professor, Dept. of ICT, SoT, PDEU

LIVE DEPLOYED CLOUD PLATFORM & API DEMO GATEWAYS

- **Public Landing Showcase:**
  https://smartserve-landing-ten.vercel.app
- **Customer Booking Portal:**
  https://smartserve-customer-alpha.vercel.app
- **Service Partner Workspace:**
  https://smartserve-provider.vercel.app
- **Admin Control Tower:**
  https://smartserve-admin-flax.vercel.app
- **Interactive Swagger API Docs:**
  https://smartserve-api-de3f.onrender.com/docs
- **API Health & Telemetry:**
  https://smartserve-api-de3f.onrender.com/health

We welcome questions, feedback, and suggestions from the Evaluation Committee.